The Energy Balance Project

Local Preservation — Version 2.9

A source-local safe-export budget, the P1C preservation controller, and its deterministic validation

Status: research note with deterministic numerical validation. NOT PEER REVIEWED. Passing tests are validation at declared fixture points, never mathematical proof. No claim of general stability, stochastic robustness, or monetary validity is made anywhere in this document.

Provenance. Every number, table, and figure below is generated at compile time from the committed machine-readable artifacts in `results/v2.9/` by `make_paper_v29.py`; the build aborts on any plan-hash or headline mismatch. Authoritative text source: `Foundation_v2.9_local_preservation.md`. Locked plan hashes: deterministic `af8f119b4af43329`…, D9/D10 `87ad0ae2eb3cca6d`….

The V2.9 question. *Can a spatially local process determine how much a regenerative source may safely export, preserve its certified reserve, and reveal infeasible demand — without evaluating the whole world or simulating its future?*

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1. Motivation and the original project objective

The Energy Balance Project asks whether **purely local physical rules** can keep a resource field alive — no central optimizer, no global objective, no simulated future. Since V2.0 the model is a scalar stock field on a graph: each cell has a viable band, cells exchange stock over lossy edges, and “health” is a quadratic burden functional that the engine never optimizes — survival must emerge or fail on its own.

The project’s stated long-term objective (V2.9 objective-alignment draft, §1) is not maximal service. It is an **honest physical sustainability signal**, in priority order: (1) preserve the regenerative foundation; (2) calculate and report the true sustainable allowance; (3) record ecological debt whenever consumption exceeds it; and only then (4)–(6) build EBU accounting and economic layers. Two guardrails bind every future layer: **no laundering by price** (cost is never permission for irreversible harm) and **no laundering by transfer** (moving tokens redistributes responsibility, never reduces planetary debt).

V2.9 is the stage where **preservation** first gets a concrete, testable, purely local mechanism, and the **sustainable allowance** gets its first formula: a safe-export budget a source computes from its own frozen state. The deterministic answer, in the fixtures tested: **yes for one-step reserve preservation and honest rationing** — with sharp, recorded limits: feasibility is separate (no export rule can create it), and nothing here proves long-horizon or stochastic viability (§15–16).

2. What was known at V2.8

V2.7 proved a continuous-time energy–dissipation identity for the idealized Onsager transport law: along solutions,

$$\frac{dV}{dt} \;=\; \sum_i \mu_i u_i \;-\; \sum_e \left(\frac{J_e^2}{M_e} + \theta_e J_e\right),$$

where V is the total burden (per-cell quadratic penalties), $\mu_i = \partial V/\partial x_i$ the local marginal, u_i the local natural drive (supply + regeneration – demand – leakage), and each edge e (mobility M_e, threshold θ_e, efficiency η_e) carries the loss-aware flux $J_e = M_e[\mu_i - \eta_e\mu_j - \theta_e]_+$. Transport can only dissipate; only drive can raise the burden.

V2.8 crossed half the continuous→discrete gap. For **Model D0** — the synchronous, frozen-state, **unconstrained**, loss-aware explicit-Euler law

$$x^{n+1} \;=\; x^n + \Delta t\,(u(x^n) + SJ(x^n))$$

— it proved the finite-step burden inequality (V2.8 Thm 4.4/6.1):

$$V(x^{n+1}) - V(x^n) \;\le\; \Delta t\,\mu^\top u \;-\; \Delta t \sum_e \left(\frac{J_e^2}{M_e} + \theta_e J_e\right) \;+\; \frac{L_V \Delta t^2}{2}\,\|u + SJ\|^2,$$

drive minus dissipation plus an explicit step-size penalty, with a corrected curvature constant L_V that **sums** simultaneously active penalty weights. Its §11 scope list excludes clipping, projection, spill, saturation, hard-reserve constraints, and sequential updates: **any constrained flow is outside the V2.8 theorem**. That exclusion is exactly where V2.9 begins — a preservation controller must constrain flows, so it must bring its own mathematics.

3. Why soft penalties are not hard preservation

The burden can include a reserve term $\chi_i[R_i - x_i]_+^2$ that penalizes dipping below a reserve. It shapes the force: as the source falls toward R, the reserve marginal $-2\chi(R - x)_+$ pushes against export. But it is **soft**, and the algebra says why it cannot guarantee anything:

> **Observation (alignment draft 5.1).** The reserve penalty and its derivative both vanish at x = R itself. At the boundary there is no force left; one more finite tick of demand steps straight through. A soft penalty can *delay* damage — in D9 it measurably does (§10) — but it cannot *forbid* it. Hard preservation needs a constraint, not a gradient.

4. The safe-export budget

Freeze one source cell at tick start: stock x, local drive u, timestep Δt, certified effective reserve R^{eff}. If the source exports an aggregate rate Q for one tick (ignoring incoming flow, which only helps), its successor is $x' = x + \Delta t(u - Q)$. Requiring $x' \geq R^{eff}$ and solving for Q gives the largest reserve-safe aggregate export rate; with explicit physical measurement margins ε_x (stock) and ε_u (drive) it becomes the **robust budget** (Amendment 4, A4.5):

$$Q_{max} = \frac{[x + \Delta t\, u - R^{eff}]_+}{\Delta t}, \qquad Q^{rob}_{max} = \frac{[(x - \varepsilon_x) + \Delta t\,(u - \varepsilon_u) - R^{eff}]_+}{\Delta t}.$$

Every symbol is local: the source's own stock, drive, reserve certificate, and the tick length. No global state, no other cell, no future rollout. Two structural points (both from the independent Gate-2.1B review, which corrected the first design): the budget governs **the sum, not each edge** (else k edges could export $k \cdot Q_{max}$); and **feasibility is a separate hypothesis** — $Q_{max} = 0$ means "no export allowed", not "safe".

> **Feasibility counterexample (review Obs 4.2).** With g ≡ 0, u = –δ < 0 and $x = R^{eff}$, even zero export gives $x' = R^{eff} - \Delta t\delta < R^{eff}$. When natural decline alone crosses the reserve, no export rule can preserve it; the controller must *report* that state, never claim to have prevented it.

This yields the action-time state classification, computed from frozen local data only (`p1c_v29.classify_state`):

State	Condition	Export
P (preservable)	$x \geq R^{eff}$ and $x + \Delta t \cdot u \geq R^{eff}$	up to Q_{max}^{rob} (regenerative)
R (recovery required)	$x < R^{eff}$	zero
I (locally infeasible)	$x \geq R^{eff}$ but $x + \Delta t \cdot u < R^{eff}$	zero — and zero cannot save the boundary; reported explicitly
F (flow source)	external renewable flow	declared flow cap; no stock reserve

Finite and irreversible stocks get a **zero** preservation-safe budget even in state P: positive extraction of a non-regenerating stock is depletion, not preservation.

5. Aggregate multi-edge allocation

When a source's edges request more than its budget — aggregate raw request $Q_{req} = \Sigma_e\, q_e^{req}$ exceeding Q_{max}^{rob} — P1C scales **all** of that source's outgoing requests by one factor computed from frozen state:

$$\sigma = \min\left(1,\ \frac{Q^{rob}_{max}}{Q_{req}}\right), \qquad q_e^{acc} = \sigma\, q_e^{req}.$$

Proportional scaling keeps the decision local (one number per source), treats edges symmetrically, and composes with the synchronous update without ordering artifacts. Raw requests are the unmodified loss-aware D0 fluxes (reused, not forked). Uncommitted **incoming** flow is excluded from the budget: a source must stay safe even if every upstream neighbor withholds this tick; delivered incoming flow appears only as a diagnostic.

6. The P1C tick sequence

One P1C tick (`p1c_v29.p1c_step`), in frozen order:

Step	Action
1	**Freeze** the state vector; compute each cell's local drive u_i from its own frozen stock (D0 semantics, unchanged).
2	**Gather** raw loss-aware edge requests q_e^{req} from frozen local views (the D0 local law, unchanged).
3	**Classify** each configured source (P/R/I/F) and compute its ONE aggregate safe-export budget (§4).
4	**Scale** that source's requests proportionally when they exceed the budget (§5).
5	**Apply** all accepted flows in one simultaneous update: the source loses the full withdrawal q_e^{acc}; the destination receives only $\eta_e\, q_e^{acc}$.
6	**Count service after transport loss**: delivered service is $\eta_e\, q_e^{acc}$, never the pre-loss withdrawal.
7	**Report unmet demand explicitly**: the requested–delivered gap is a first-class output, never renamed, never hidden.

$$x_i^{n+1} = x_i + \Delta t\left(u_i + \sum_{e \in \mathrm{in}(i)} \eta_e q_e^{acc} - \sum_{e \in \mathrm{out}(i)} q_e^{acc}\right)$$

Everything on the decision path reads frozen local data only. The global functional V is computed *after* the update by the researcher harness as a diagnostic — never by the controller (enforced by AST tests).

7. The one-step preservation result

Theorem (one-step aggregate reserve preservation; Gate-2.1B review, Thm 4.1). Fix a tick. Suppose a protected regenerative source i satisfies, on its frozen state: (1) $x_i \ge R_i^{eff}$; (2) $x_i + \Delta t{\cdot}u_i \ge R_i^{eff}$ (its no-export successor remains feasible — state P); and (3) its aggregate accepted export satisfies $Q_i^{acc} \le Q_{i,max}$, where

$$Q_{i,\max} = \frac{\left[x_i + \Delta t\, u_i - R_i^{\mathrm{eff}}\right]_+}{\Delta t},$$

and the synchronous update of §6 step 5 is applied. Then $x_i^{n+1} \ge R_i^{eff}$.

Proof. Incoming flow is non-negative, so $x_i^{n+1} \ge x_i + \Delta t{\cdot}u_i - \Delta t{\cdot}Q_i^{acc}$. By (3), $\Delta t{\cdot}Q_i^{acc} \le [x_i + \Delta t\, u_i - R_i^{eff}]_+$, which under (2) equals $x_i + \Delta t\, u_i - R_i^{eff}$. Hence $x_i^{n+1} \ge R_i^{eff}$. ■

With the robust budget the same algebra holds whenever the true stock and drive lie within the declared margins ε_x, ε_u. P1C satisfies hypothesis (3) by construction ($\sigma \le Q_{max}^{rob}/Q_{req}$), so every state-P tick is an instance of the theorem; the harness records per-tick conformance.

What this theorem is not. A one-step algebraic statement under explicit hypotheses. It is NOT: an infinite-horizon sustainability theorem; a proof of global homeostasis; a proof for arbitrary (asynchronous, sequential) actor scheduling; a proof under model uncertainty beyond the declared ε margins; a proof of complete service (P1C rations precisely when demand is unsafe); or an EBU accounting theorem (no EBU exists in V2.9). Feasibility (hypothesis 2) is an assumption, not a conclusion.

Relationship to V2.8

The V2.8 finite-step inequality holds for the **unconstrained** D0 flux; capping the aggregate export changes the flux, so that theorem does **not** transfer to P1C (every P1C tick record carries `covered_by_v28_theorem = False`). P1C carries its own one-step reserve algebra (above) plus numerical conformance checks (`test_v29_p1c.py`, 83 checks). The **joint** theorem — constrained preservation *and* a dissipation/descent inequality for the capped flux — remains open (§16).

8. Feasibility: recovery-required and infeasible states

The P/R/I/F classifier makes the negative cases first-class outputs. **State R** ($x < R^{eff}$): the reserve is already breached; export is zero; recovery is up to the source's own regeneration — P1C makes no recovery claim. **State I** ($x \geq R^{eff}$ but $x + \Delta t \cdot u < R^{eff}$): natural decline alone will cross the reserve this tick; export is zero AND the controller reports `zero_export_insufficient = true`. This is the honest-signal case: the demand structure is infeasible for this source and no local export rule can fix it — the information must go up to whatever layer sets demand.

D10's demand axis deliberately crosses both analytic boundaries: $d/g_{max} = 1$ is the logistic fold (maximum sustainable regeneration), and delivering demand d through an edge of efficiency η requires $d \leq \eta \cdot g_{max}$. Points with $\eta < d/g_{max} < 1$ are regeneration-feasible but transport-infeasible: the committed map (§11) shows P1C rationing exactly there.

9. D1–D8: deterministic validation of the substrate

Before any preservation experiment, V2.9 Gate 2 ran a preregistered deterministic wind tunnel (plan hash `af8f119b4af43329`…, 24 runs, fixtures D1–D8) on the frozen D0 engine `d0_v29.py`, comparing P0 (no transport), P1 (exact D0), P1K (diagnostic projection wrapper), and ablations P2 (loss-blind force), P3 (sequential live-state), P4 (oversized step — deliberately unsafe). Headline committed outcomes:

Fixture	Question	Committed outcome
D1	lossless relaxation	P1 drives V 17.0 → 6.3×10^{-30}, zero descent violations; P0 exactly constant.
D2	lossy transport vs loss-blind ablation	P1 rests where the loss-aware force ≈ θ (0.0500 vs θ = 0.05) while the loss-blind force is still 5.29; final V: P1 58.24 < P2 61.81 (11 descent violations recorded for P2).
D3	one-tick causality	synchronous P1: probe difference 0.0 at tick 1, first difference at tick 2; sequential P3 leaks 0.0162 at tick 1.
D4/D6/D7	driven service + shocks	P1 serves 100% of feasible demand from admissible stock and recovers after supply/demand shocks (recovery ticks 440 and 611); P0 exits the physical domain.
D5	Allee reserve pressure	**did NOT discriminate**: penalty ON and OFF arms identical on headline outcomes (both converged, both served 150/150, no Allee crossings). Retained; motivated D9.
D8	oversized-step negative control	tick-1 overshoot ratio exactly 1.96 as registered; the monotone-growth sub-claim FAILED (recorded, not repaired). Certified contrast non-increasing.

Interpretation limits are frozen in the plan: deterministic fixtures only; a success is evidence the law *can* work in that fixture, never that it always works. P1K is a diagnostic projection wrapper whose ledger closes by construction — closure never certifies that delivered stock was physically available, so P1K can never found a physical-service claim.

10. D9: the Allee reserve-stress experiment

D9 (four arms, 200 ticks, $\Delta t = 0.2$) puts an Allee source ($A = 5$, $K = 20$, certified $R^{eff} = 11$) under sustained demand 5.0 — deliberately hard enough that unconstrained transport breaches the reserve. All values below are read from the committed summary at build time.

Arm	Policy	Class	Reserve crossed	Allee crossed	O_physical	Delivered	Unmet
D9-A	P1 reserve-blind	collapse	tick 8	tick 17	201.845	189.69	10.31
D9-B	soft (χ=1)	collapse	tick 8	tick 24	183.045	172.83	27.17
D9-C	P1C (χ=1, cap)	safe rationing	never	never	$\le 10^{-9}$	130.82	69.18
D9-D	P1C (χ=0, cap)	safe rationing	never	never	$\le 10^{-9}$	130.82	69.18

Table: the four committed D9 arms (results/v2.9/d9_d10). P1C binding on 193/200 ticks; 200 theorem-eligible ticks with 0 observed violations.

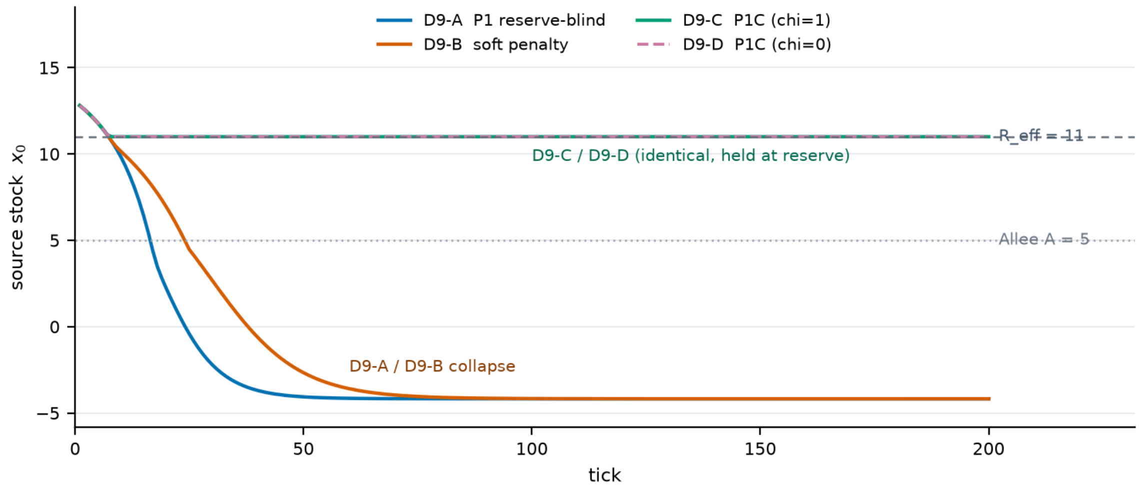

Figure: committed D9 source-stock trajectories (from the per-tick trace). Reserve-blind (D9-A) and soft (D9-B) cross R^{eff} at tick 8 and the Allee threshold at ticks 17 and 24; both P1C arms hold the source at exactly $R^{eff} = 11$ for all 200 ticks.

Readings. The soft penalty **delayed** the Allee crossing (tick 24 vs 17) and reduced over-use (183.0 vs 201.8) but crossed the reserve at the **same tick 8**: it slows damage, it does not prevent it (§3). Both P1C arms held the source at exactly $R^{eff} = 11$ (final = minimum = 11.0) with zero reserve and zero Allee crossings, the cap binding on 193/200 ticks, and **zero observed one-step-preservation violations over 200 eligible ticks**. P1C delivered less (130.8 vs 189.7) and reported the difference honestly as unmet demand (69.2): the demand was physically unsafe, and rationing — not silent depletion — is the designed behavior. **D9-C and D9-D are identical on every recorded metric: the hard cap, not χ, provides preservation in this fixture.**

11. D10: the service-versus-preservation phase map

D10 sweeps a logistic source–sink world (140 runs: an 80-run core map over demand ratio $d/g_{max} \in \{0.25, 0.5, 0.9, 1.0, 1.1\}$ × transport efficiency $\eta \in \{0.5, 0.7, 0.9, 1.0\}$, plus 60 secondary-slice runs over θ, δ/K, χ, ρ, r_{dt}), four policies at every point. Committed classification counts (35 runs per policy):

Policy	collapse	safe_service	safe_rationing	runs with O_physical > 10^{-9}
P0	0	0	35	0
P1	25	10	0	25
soft	25	10	0	25
P1C	0	25	10	0

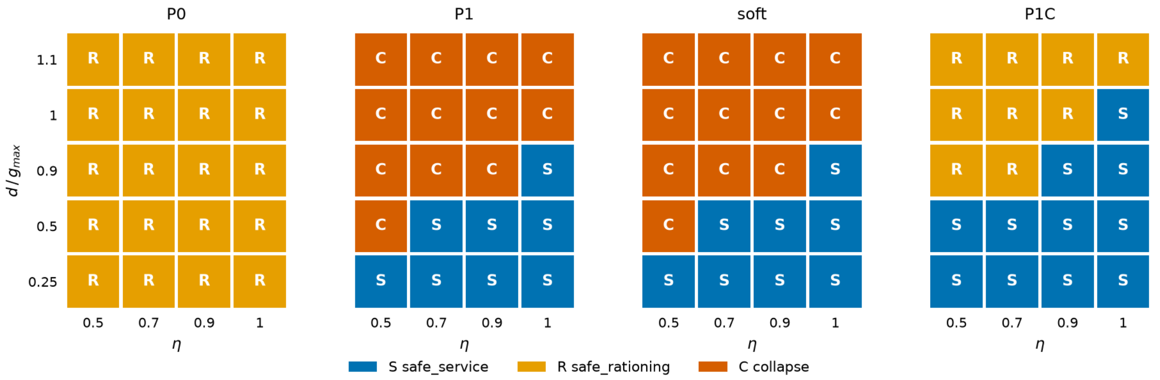

Figure: committed D10 core phase map (classification per policy per grid point; S = safe_service, R = safe_rationing, C = collapse), generated from the summary JSON.

P1 and the soft arm produce **identical** classification maps: at these demand levels the soft penalty changes nothing that matters. Collapse for P1/soft tracks the registered analytic boundaries — every core point with $d/g_{max} \geq 0.9$ collapses except at $\eta = 1.0$ with $d/g_{max} \leq 0.9$, plus the transport-starved corner (0.5, 0.5); safe service requires roughly $d/g_{max} \leq \eta$ (deliverability) and $d/g_{max} < 1$ (the fold). P1C never collapses and never over-uses; where full service is deliverable it serves (25 points), where it is not it rations (10 points). Hypothesis H-D10-1 held at all tested points. 18 runs (P1/soft at the hardest corners) diverged and exited the physical domain; all 18 domain exits are **recorded, not dropped**.

> **P0's 35 "safe rationing" runs are not success.** No transport trivially preserves stock by serving nothing. Preservation must always be read *together with service*: P1C matches P1's service wherever service was safe, and refuses only the unsafe remainder.

12. Serialization provenance (Attempt 1, Attempt 2, the repair)

The D9/D10 study has a fully documented execution history. **Attempt 1 failed** (harness-integrity defect found by Gate 2.4A before any scientific use) and is preserved verbatim (`ATTEMPT_1_FAILURE.md`); it produced no scientific result. **Attempt 2 ran exactly once** (implementation commit `12faa53`…, plan hash unchanged) and produced all 144 registered runs. **No physical trajectory was ever regenerated after that run.**

On 15 diverging runs the two aggregate stability diagnostics (`stability_tau`, `stability_amp`) overflowed to infinity; under explicit authorization they were post-processed to JSON null — values only, no classification, trajectory, or other metric changed. Three further domain-exit records were **natively undefined** (exit before the 100-tick burn-in left an empty diagnostic window), so the committed summary contains 18 null diagnostic pairs: 15 repairs + 3 native (`ATTEMPT_2_SERIALIZATION_REPAIR.md`). Gate 2.4B then made future serialization **fail closed**: strict JSON (allow_nan=False) everywhere, one narrowly scoped normalization for the two nullable diagnostics (with recorded reasons), and a hard error for a non-finite value in any state, flow, service, unmet-demand, crossing, classification, over-use, ledger, or timestep field (`serialization_v29.py`, 46 checks).

13. What failed, and why it is kept

Negative and partial results are retained as first-class outputs: **D5 failed to discriminate** (the fixture was too easy; D9 was designed from that failure). **The soft penalty failed in D9/D10** exactly as the algebra predicts: delay, not prevention. **Unconstrained P1 collapses and over-uses** across 25/35 of the D10 map — the honest baseline preservation must beat. **D8's monotone-growth sub-claim failed** (overshoot fired exactly at tick 1 but did not keep growing). **Attempt 1 of D9/D10 failed** on harness integrity and is preserved. **The stability diagnostics overflowed** on diverging runs — a serialization defect, fixed fail-closed (§12). Inherited from V2.6 and still open: the guarded EBU ledger is **exploitable on at least one topology** (+260 EBU while every source dies); any future credit rule inherits that risk until re-audited.

14. Relationship to future EBU accounting

V2.9 is a **physical controller layer**, deliberately below any accounting. The objective-alignment documents design — but do **not** implement — the next layers: a **vector** ecological-debt ledger $D = (D_{carbon}, D_{water}, \ldots)$ with an irreversible component that never decreases and no netting between accounts; one-to-one EBU **restoration credit** only for verified net physical restoration; and the guardrails G1/G2 (§1). The independent review's verdict on that design is "pass with corrections", and it explicitly finds **scalar (fungible) EBU not currently justified** — a first implementation would use resource-specific vector accounts.

> **None of this exists in code.** V2.9 contains no EBU issuance, no ecological-debt ledger, no wallet, no exchange, no scalarisation, no actor health or death, and no finite moving actor population. `O_physical` is a physical over-use *diagnostic*, not issued debt.

15. Limitations and falsification criteria

Limitations (read before citing any result). Deterministic toy worlds only (2–3 cells); no stochastic or confirmatory seed study for P1C yet; no random layouts. No finite moving actor population; no individual health or death; no complete society. No EBU issuance, debt ledger, wallets, exchange, or scalarisation. The preservation theorem is one-step, synchronous, and margin-conditional; reserve certification (R^{eff}) is assumed given, and long-horizon viability — the infinite-horizon kernel — remains open. No proof under arbitrary uncertainty: the ε margins cover declared measurement error only, not model mis-specification. Not peer reviewed. All checks are numerical validation at declared points, never proof.

Falsification criteria (registered in the alignment draft; fixture falsifiers in the plans). The architecture is falsified by, among others: (F-A) a trajectory labeled safe that in fact crosses a non-substitutable reserve or increases irreversible debt; (F-B) a safe-export certificate that, applied repeatedly under admissible drive, still drives a source below its reserve; (F-C) wallet transfers that reduce recorded physical debt; (F-D) damage-then-repair earning net positive credit without ending above the pre-damage baseline. A certificate architecture that cannot emit UNSUSTAINABLE/INFEASIBLE at all is rejected by construction. Within V2.9's committed studies, the registered falsifiers (e.g. F-D10-1: P1C collapsing while theorem assumptions hold) were **not** triggered.

16. Open mathematical and computational problems

#	Problem
1	**Joint constrained theorem**: preservation PLUS a descent/dissipation inequality for the capped flux (the V2.8 analogue for P1C).
2	**Infinite-horizon controlled-invariant kernel** K_∞ and multi-step feasibility: when does one-step preservation compose forever?
3	Discrete overshoot margin for the reserve certificate (δ, ε sizing) and V2.7 Conjecture C-2 (reserve surrogate).
4	Multi-source, multi-edge joint invariance under proportional allocation (shared sinks, competing sources).
5	Stochastic and adversarial robustness of P1C (confirmatory seed study).

6	Ecological-debt attribution (overdraw → account map); an operational “verified restoration” predicate; multi-actor credit attribution; scalarisation weights — prerequisites for any EBU layer.
7	Reserve certification itself: who computes R^{eff}, how often, with what non-local ecological model (the three-timescale design in the review is a sketch, not a result).

17. Reproducibility

Everything is deterministic; no seeds, no tuning knobs, no options. Validation suites (standard library, directly executable): `test_v29.py` (D0 conformance, 141 checks / 15 groups), `test_v29_behavior.py` (D1–D8, 108 / 9), `test_v29_p1c.py` (P1C conformance, 83 / 12), `test_v29_d9_d10.py` (D9/D10 harness, 114 / 20), `test_v29_serialization.py` (strict serialization, 46 / 6). The committed studies are locked: `exp_v29.py` and `exp_v29_d9_d10.py` refuse to overwrite completed results, recompute their canonical plan hashes at run time, and take no options. Committed artifacts: `results/v2.9/deterministic/` (24 runs) and `results/v2.9/d9_d10/` (144 runs, gzip trace, manifest, attempt history, serialization repair audit). Python 3.14.2 was used for the committed runs. This PDF regenerates with `venv/bin/python make_paper_v29.py`.

This note has not been peer reviewed. The one-step preservation theorem is an algebraic result under explicit hypotheses; every experimental statement is a deterministic observation at registered fixture points; the numerical suites validate, they never prove. No EBU, wallet, debt-ledger, or actor-health code exists in V2.9, and no general sustainability claim is made.